Machine Learning Prediction of Heating and Cooling Loads of Residential Buildings

Project Plan

Oğuz Kağan Göçmez

***Abstract*—Heating and cooling loads determine the size of the heating, ventilation and air conditioning (HVAC) equipment that a building requires. They are usually computed with detailed simulation software, which is time-consuming and requires expert users. Machine learning (ML) models trained on simulated buildings can provide comparable estimates in a fraction of a second. This project uses the UCI Energy Efficiency dataset, which contains 768 residential buildings simulated in Ecotect and described by eight design parameters (relative compactness, surface area, wall area, roof area, overall height, orientation, glazing area and glazing area distribution). Two outputs are predicted: the heating load (HL) and the cooling load (CL). Following the reference study of Tsanas and Xifara, we will analyze the data statistically, compare six regression models using repeated 10-fold cross-validation and three error measures, tune the most promising models, and identify the most important inputs.**

As an additional contribution, we will evaluate how well the models generalize to building shapes that are not seen during training. Buildings that differ only in orientation mostly have very similar loads. In a random split, such buildings appear in both the training and the test sets, which may lead to optimistic error estimates.

***Index Terms*—Building energy performance, heating load, cooling load, machine learning, regression, cross-validation.**

I. Introduction

HVAC systems account for most of the energy used in buildings, so estimating heating and cooling loads in the early design phase supports the design of energy-efficient buildings [1]. Simulation tools provide reliable estimates but are time-consuming, which has motivated the use of ML models such as decision trees [2], support vector machines [3] and polynomial regression [4].

Tsanas and Xifara [1] introduced the dataset used in this project [5]. They compared a robust linear regression model fitted with iteratively reweighted least squares (IRLS) and a random forest (RF), and reported mean absolute errors (MAE) of 0.51 for HL and 1.42 for CL with the RF. Later studies applied other artificial intelligence techniques [6] and genetic programming [7] to the same data. The dataset is also widely used as a benchmark: Žegklitz and Pošík [8] used it to compare symbolic regression methods with a tuned RF and support vector regression (SVR), and Yazici *et al.* [9] used it to test ML models on a low-power Internet of Things device. The project follows the Machine Learning Project Checklist of the course textbook [10].

The author is with Özyeğin University, İstanbul, Türkiye. This plan is part of the EE 311 Introduction to Artificial Intelligence, Machine Learning and Data Science course project, Fall 2026.

TABLE I
Project Schedule

Subtask	Week
Literature search	1–2
Frame the problem	1
Get the data	1
Explore the data	2
Project plan report	2
Prepare the data	3
Shortlist promising models	3–4
Grouped CV by building shape	5
Midterm report	6
Fine-tune the system	7–8
Feature importance and errors	9
Present the results	10
Final report	11–12
Recorded presentation	12

Week 1 is the week the topic was selected; week 12 is the last week of the semester.

II. Project Schedule

Table I lists the subtasks and their planned completion weeks. Since this is a single-author project, the author is responsible for all subtasks. The approach to each subtask is described in the following subsections.

A. Literature Search

We will review the reference study [1] and its references, and search for subsequent studies on the same dataset (e.g., papers citing [1] on Google Scholar) to collect results for comparison. This search will continue until the final report.

B. Frame the Problem

The task is a supervised regression problem with two continuous outputs, HL and CL. To ensure direct comparability with [1], we use the same three error measures: mean absolute error (MAE), mean squared error (MSE), and mean relative error (MRE). The IRLS and RF results reported in [1] serve as the baseline.

C. Get the Data

The dataset [5] was loaded from a CSV export of the original Excel file, which uses semicolons as separators and

commas as decimal points. The CSV file was verified against the Excel file: it contains 768 buildings and 10 columns, with no missing values and no duplicate rows.

D. Explore the Data

Histograms and input–output scatter plots were plotted, and the association between each input and each output was measured using mutual information and the Spearman rank correlation (with p-values), as in [1]. The Spearman correlation was chosen over the Pearson correlation because the variables are not normally distributed.

E. Prepare the Data

The dataset requires no cleaning. The inputs will be standardized within a pipeline so that the scaler is fitted on the training folds only. We will also evaluate one-hot encoding of orientation and glazing area distribution, as well as the removal of the redundant surface area input.

F. Shortlist Promising Models

We will compare six models implemented in scikit-learn [11]: linear regression, robust linear regression with the Huber loss (a robust alternative comparable to IRLS), k-nearest neighbors, support vector regression, random forest [12], and gradient boosting [13]. The models will be evaluated using repeated 10-fold cross-validation (CV) on random splits, as in [1]. As an additional contribution, the evaluation will be repeated with grouped CV [14], in which all buildings of the same shape (one of 12) are kept in the same fold, in order to measure the performance on unseen shapes.

G. Fine-Tune the System

For the two or three best-performing models, the hyperparameters will be tuned using grid or randomized search within CV. The final results will be obtained with 100 CV repetitions, as in [1]. In addition, we will compute the permutation importance and add a random probe feature, as in [1], to determine the most important inputs. We will also evaluate a single multi-output model that predicts both loads simultaneously.

H. Present the Results

The results will be summarized in tables and figures and compared with those of [1], [6], [7], and [8]. The midterm report, the final report, and a recorded presentation of 10–15 minutes will then be prepared.

Acknowledgment

An AI tool (Claude, Anthropic) was used in two ways in this project. First, it suggested methods, namely the use of the Spearman correlation for the non-normally distributed data and a list of candidate regression models. The author verified these suggestions against the literature and decided which ones to adopt. Second, it drafted the text of this plan based on the author's decisions, schedule, and references, and assisted with language editing. The analysis code was written by the author. All AI suggestions were carefully reviewed by the author, and their correctness was validated.

References

[1] A. Tsanas and A. Xifara, "Accurate quantitative estimation of energy performance of residential buildings using statistical machine learning tools," *Energy Build.*, vol. 49, pp. 560–567, 2012, doi: 10.1016/j.enbuild.2012.03.003.

[2] Z. Yu, F. Haghighat, B. C. M. Fung, and H. Yoshino, "A decision tree method for building energy demand modeling," *Energy Build.*, vol. 42, no. 10, pp. 1637–1646, 2010, doi: 10.1016/j.enbuild.2010.04.006.

[3] B. Dong, C. Cao, and S. E. Lee, "Applying support vector machines to predict building energy consumption in tropical region," *Energy Build.*, vol. 37, no. 5, pp. 545–553, 2005, doi: 10.1016/j.enbuild.2004.09.009.

[4] T. Catalina, J. Virgone, and E. Blanco, "Development and validation of regression models to predict monthly heating demand for residential buildings," *Energy Build.*, vol. 40, no. 10, pp. 1825–1832, 2008, doi: 10.1016/j.enbuild.2008.04.001.

[5] A. Tsanas and A. Xifara, "Energy Efficiency," UCI Machine Learning Repository, 2012. [Online]. Available: https://doi.org/10.24432/C51307

[6] J.-S. Chou and D.-K. Bui, "Modeling heating and cooling loads by artificial intelligence for energy-efficient building design," *Energy Build.*, vol. 82, pp. 437–446, 2014, doi: 10.1016/j.enbuild.2014.07.036.

[7] M. Castelli, L. Trujillo, L. Vanneschi, and A. Popovič, "Prediction of energy performance of residential buildings: A genetic programming approach," *Energy Build.*, vol. 102, pp. 67–74, 2015, doi: 10.1016/j.enbuild.2015.05.013.

[8] J. Žegklitz and P. Pošík, "Symbolic regression algorithms with built-in linear regression," 2017, arXiv:1701.03641. [Online]. Available: https://arxiv.org/abs/1701.03641

[9] M. T. Yazici, S. Basurra, and M. M. Gaber, "Edge machine learning: Enabling smart Internet of Things applications," *Big Data Cogn. Comput.*, vol. 2, no. 3, Art. no. 26, 2018, doi: 10.3390/bdcc2030026.

[10] A. Géron, *Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow*, 3rd ed. Sebastopol, CA, USA: O'Reilly Media, 2022.

[11] F. Pedregosa *et al.*, "Scikit-learn: Machine learning in Python," *J. Mach. Learn. Res.*, vol. 12, pp. 2825–2830, 2011. [Online]. Available: https://jmlr.org/papers/v12/pedregosa11a.html

[12] L. Breiman, "Random forests," *Mach. Learn.*, vol. 45, no. 1, pp. 5–32, 2001, doi: 10.1023/A:1010933404324.

[13] J. H. Friedman, "Greedy function approximation: A gradient boosting machine," *Ann. Statist.*, vol. 29, no. 5, pp. 1189–1232, 2001, doi: 10.1214/aos/1013203451.

[14] D. R. Roberts *et al.*, "Cross-validation strategies for data with temporal, spatial, hierarchical, or phylogenetic structure," *Ecography*, vol. 40, no. 8, pp. 913–929, 2017, doi: 10.1111/ecog.02881.